Straightening Laws in Algebraic Complexity

Anakin Dey

What is Algebraic Complexity?

Question

How hard is it to compute polynomials?

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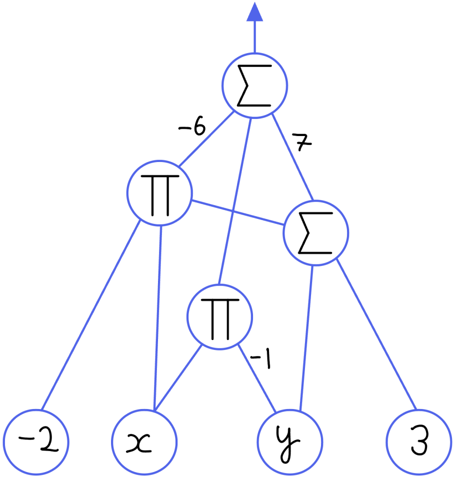
Can we find *explicit* polynomials which are hard to compute?

Algebraic Circuits



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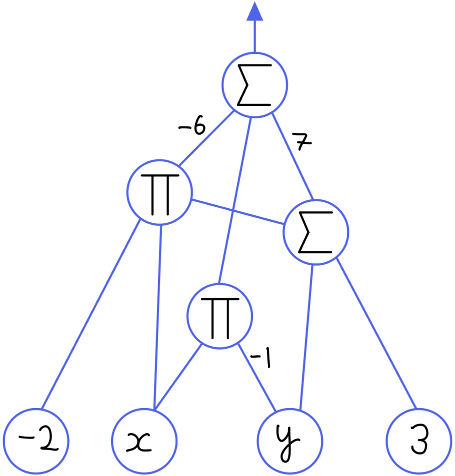

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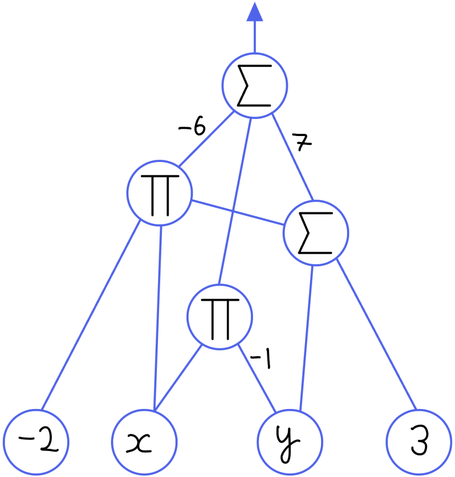

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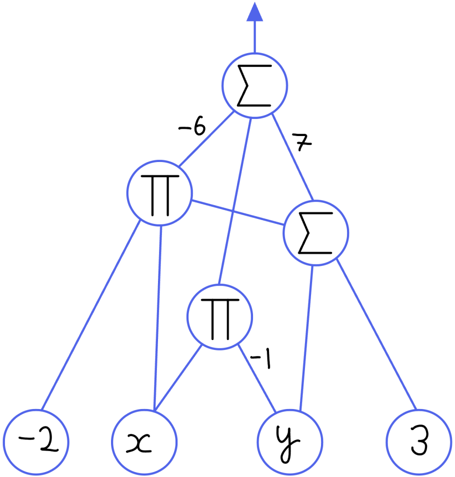

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- *Size:* The number of edges
- *Depth:* The length of the longest input-output path

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- The number of additions is optimal [Ost54].
- The number of multiplications is optimal [Pan66].

Hard Polynomials Exist

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Let $\mathbb{F}$ be a field of characteristic zero. Then "most" n-variate polynomials of degree $\leq d$ require circuits of size $N := \binom{n+d}{d}$.

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- $\dim \operatorname{im} \Phi \leq s$, and there are only finitely many unlabeled circuits.
- Thus, the dimension over all possible unlabeled circuits is at most s.
- For the image to be Zariski-dense, we must have that $s \geq N$.

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Then m must be *superpolynomial* in n.

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We say a circuit *border computes* $g(\overline{x})$ over $\mathbb{F}(\varepsilon)$ if it computes

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We cannot set $\varepsilon = 0$ since we may divide by ε.

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What is the power of border computation?

Is it strictly necessary? Can we prove something in the border setting and *deborder* the result to get an exact computation?

Border Waring Rank

Definition

The *Waring Rank*, $\mathrm{WR}(f)$, of a degree d form $f(\overline{x})$ over $\mathbb{C}$ is the minimum r such that there exist linear forms $\ell_1(\overline{x}), \ldots, \ell_r(\overline{x})$ such that

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$\implies \{ f(x, y) \in \mathbb{C}[x, y]_{\leq d} \mid \mathrm{WR}(f) = 2 \}$ is *not* Zariski-closed.

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Suppose $f \in \langle g_1, \ldots, g_k \rangle$. How does the complexity of f compare to the complexity of the generators $g_1, \ldots, g_k$?

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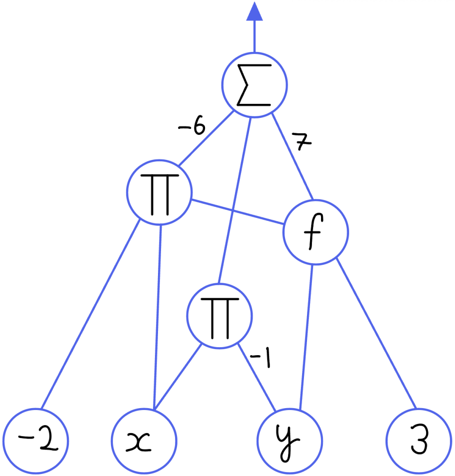

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How can we compare the *relative* complexity of two polynomials?

We can allow a fixed polynomial $f(\overline{x})$ to be computed with unit cost by allowing it to function as its own gate.

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Suppose $0 \neq f \in \langle g \rangle$. Does g have a small f-oracle circuit?

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This is open even if we ask that $\{\, g_1, g_2\,\}$ are a Gröbner basis or form a regular sequence.

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- However, $\det_n = \mathrm{perm}_n + (\det_n - \mathrm{perm}_n) \in I$ is known to have a circuit of size $\mathrm{poly}(n)$ computing it.
- Thus, our question is asking us to resolve $\det_n$ versus perm_n.

Adding $\det_n$ to the set of generators remedies this issue.

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- However, these circuits are *not* constant-depth.
- The oracle allows *constant depth* computation of the determinant.

Closure Results in Determinantal Ideals

Theorem ([AF22, Theorem 1.1])

Let $f \in I^{\mathrm{det}}_{n,m,r}$ be a nonzero polynomial. Then there is a depth-three f-oracle circuit of size $O\left(n^2 m^2\right)$ that *border computes* the $t \times t$ determinant for some $t = O\left(r^{1/3}\right)$.

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Theorem ([DG25, Theorem 1.5])

Let $f \in I^{\det}_{n,m,r}$ be a nonzero polynomial. Then there is a depth-three f-oracle circuit of size $\operatorname{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = O\left(r^{1/3}\right)$.

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- We use *Straightening Laws* from Invariant Theory to express $f(X)$ in a standard basis indexed by Young tableau, and leverage the combinatorics to talk about specific terms.
- To get a circuit for a specific basis term, we use *Interpolation* as well as a novel application of the *Isolation Lemma*.

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- Working with the monomial basis is hard since determinants have exponentially many terms.

Dealing with Cancellations

- We saw earlier that we have to be careful of cancellations when working with ideals.
- Working with the monomial basis is hard since determinants have exponentially many terms.
- Thus, we will work with a different basis for $I^{\det}_{n,m,r}$ which is described combinatorially with Young tableau.

A Natural Generating Set for Determinantal Ideals

Definition (Bideterminants)

Let $\sigma = (4, 2, 1)$.

A Natural Generating Set for Determinantal Ideals

Definition (Bideterminants)

Let $\sigma = (4, 2, 1)$. Consider the *bitableau* $(S \mid T)$ of shape σ

$$(S \mid T) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 4 & 5 \\ \hline 3 & 6 & \multicolumn{2}{c}{} \\ \cline{1-2} 4 & \multicolumn{3}{c}{} \\ \cline{1-1} \end{array} \;\middle|\; \begin{array}{|c|c|c|c|} \hline 1 & 3 & 5 & 6 \\ \hline 2 & 7 & \multicolumn{2}{c}{} \\ \cline{1-2} 3 & \multicolumn{3}{c}{} \\ \cline{1-1} \end{array} \right).$$

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Then the *bideterminant* $(S \mid T)(X)$ is given by

$$(S \mid T)(X) = \begin{vmatrix} x_{1,1} & x_{1,3} & x_{1,5} & x_{1,6} \\ x_{2,1} & x_{2,3} & x_{2,5} & x_{2,6} \\ x_{4,1} & x_{4,3} & x_{4,5} & x_{4,6} \\ x_{5,1} & x_{5,3} & x_{5,5} & x_{5,6} \end{vmatrix} \begin{vmatrix} x_{3,2} & x_{3,7} \\ x_{6,2} & x_{6,7} \end{vmatrix} \left| x_{4,3} \right| .$$

A Natural Generating Set for Determinantal Ideals

Lemma

A degree d monomial $\prod_{i=1}^{d} x_{r_i,c_i}$ is given by a bideterminant:

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Thus, the bideterminants span $\mathbb{F}[X]$.

A Natural Generating Set for Determinantal Ideals

Theorem ([DRS74, §8, Theorem 3], [dCEP80, Corollary 2.3])

Let $(S \mid T)(X)$ be a bideterminant of shape σ.

A Natural Generating Set for Determinantal Ideals

Theorem ([DRS74, §8, Theorem 3], [dCEP80, Corollary 2.3])

Let $(S \mid T)(X)$ be a bideterminant of shape σ. Then $(S \mid T)(X)$ can be uniquely expressed as a linear combination

$$(S \mid T)(X) = \sum_{(A|B)} c_{A,B}(A \mid B)(X), \qquad c_{A,B} \in \mathbb{Z}$$

where A and B are *conjugate semistandard*: strictly increasing along rows, weakly increasing down columns.

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where A and B are *conjugate semistandard*: strictly increasing along rows, weakly increasing down columns.

Furthermore, if $(A \mid B)$ has shape τ, then $\tau \geq \sigma$ (*dominance order*).

A Natural Generating Set for Determinantal Ideals

Since $I^{\mathrm{det}}_{n,m,r}$ is generated by the $r \times r$ minors, the bideterminants in our basis should have at least one determinant of length at least r.

A Natural Generating Set for Determinantal Ideals

Since $I_{n,m,r}^{\mathrm{det}}$ is generated by the $r \times r$ minors, the bideterminants in our basis should have at least one determinant of length at least r.

Proposition ([BC03, Corollary 1.7])

The ideal $I_{n,m,r}^{\mathrm{det}}$ has as basis the conjugate semistandard bideterminants of shape σ whose first row has length at least r.

Shuffle Products

$$\left(\begin{array}{|c|c|c|c|c|c|}\hline i_1^{\bullet} & \cdots & i_p^{\bullet} & i_{p+1} & \cdots & i_s \\ \hline \ell_1^{\bullet} & \cdots & \ell_q^{\bullet} & \ell_{q+1} & \cdots & \ell_t \\ \hline \end{array} \;\Bigg|\; \begin{array}{|c|c|c|}\hline j_1 & \cdots & j_s \\ \hline m_1 & \cdots & m_t \\ \hline \end{array}\right)$$

$$= \sum_{\sigma} \operatorname{sgn}(\sigma) \left(\begin{array}{|c|c|c|c|c|c|}\hline \sigma(i_1) & \cdots & \sigma(i_p) & i_{p+1} & \cdots & i_s \\ \hline \sigma(\ell_1) & \cdots & \sigma(\ell_q) & \ell_{q+1} & \cdots & \ell_t \\ \hline \end{array} \;\Bigg|\; \begin{array}{|c|c|c|}\hline j_1 & \cdots & j_s \\ \hline m_1 & \cdots & m_t \\ \hline \end{array}\right)$$

Where the summation is over all permutations of $\{ i_1, \ldots, i_p, \ell_1, \ldots, \ell_q \}$ such that $\sigma(i_1) < \cdots < \sigma(i_p)$ and $\sigma(\ell_1) < \cdots < \sigma(\ell_q)$.

Laplace Expansion

$$\left(\begin{array}{|c|c|c|c|c|c|}\hline i_1 & \cdots & i_p & i_{p+1} & \cdots & i_s \\ \hline \ell_1 & \cdots & \ell_q & \ell_{q+1} & \cdots & \ell_t \\ \hline\end{array} \;\middle|\; \begin{array}{|c|c|c|}\hline j_1 & \cdots & j_s \\ \hline m_1 & \cdots & m_t \\ \hline\end{array}\right)$$

$$= \sum_{\tau,\mu} \operatorname{sgn}(\tau)\operatorname{sgn}(\mu) \left(\begin{array}{|c|c|c|c|c|c|}\hline i_1 & \cdots & i_p & \ell_1 & \cdots & \ell_q \\ \hline i_{p+1} & \cdots & i_s & \multicolumn{3}{c}{} \\ \cline{1-3} \ell_{q+1} & \cdots & \ell_t & \multicolumn{3}{c}{} \\ \cline{1-3}\end{array} \;\middle|\; \begin{array}{|c|c|c|c|c|c|}\hline \tau(j_1) & \cdots & \tau(j_p) & \mu(m_1) & \cdots & \mu(m_q) \\ \hline \tau(j_{p+1}) & \cdots & \tau(j_s) & \multicolumn{3}{c}{} \\ \cline{1-3} \mu(m_{q+1}) & \cdots & \mu(m_t) & \multicolumn{3}{c}{} \\ \cline{1-3}\end{array}\right)$$

Where the summation is over all permutations of $\{ j_1, \ldots, j_p, j_{p+1}, \ldots, j_s \}$ such that $\tau(j_1) < \cdots < \tau(j_p)$ and $\tau(j_{p+1}) < \cdots < \tau(j_s)$ and similar for μ.

Example Straightening

Consider the following nonstandard tableau

$$\left(\begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & 4 \\ \hline 2 \\ \cline{1-1} \end{array} \;\middle|\; \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 1 & 3 \\ \hline 1 \\ \cline{1-1} \end{array} \right)$$

Example Straightening

Shuffle:

$$\left(\ \begin{array}{|c|c|}\hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array}\ \middle|\ \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array}\ \right) = \left(\ \begin{array}{|c|c|}\hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array}\ \middle|\ \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array}\ \right) - \left(\ \begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 & 4 \\ \hline \end{array}\ \middle|\ \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array}\ \right) + \left(\ \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array}\ \middle|\ \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array}\ \right)$$

Example Straightening

Shuffle:

$$\left(\begin{array}{|c|c|}\hline 2&3\\\hline 1&4\\\hline\end{array}\;\middle|\;\begin{array}{|c|c|}\hline 1&2\\\hline 1&3\\\hline\end{array}\right)=\left(\begin{array}{|c|c|}\hline 2&3\\\hline 1&4\\\hline\end{array}\;\middle|\;\begin{array}{|c|c|}\hline 1&2\\\hline 1&3\\\hline\end{array}\right)-\left(\begin{array}{|c|c|}\hline 1&3\\\hline 2&4\\\hline\end{array}\;\middle|\;\begin{array}{|c|c|}\hline 1&2\\\hline 1&3\\\hline\end{array}\right)+\left(\begin{array}{|c|c|}\hline 1&2\\\hline 3&4\\\hline\end{array}\;\middle|\;\begin{array}{|c|c|}\hline 1&2\\\hline 1&3\\\hline\end{array}\right)$$

Laplace:

$$\begin{aligned}\left(\begin{array}{|c|c|}\hline 2&3\\\hline 1&4\\\hline\end{array}\;\middle|\;\begin{array}{|c|c|}\hline 1&2\\\hline 1&3\\\hline\end{array}\right)&=\left(\begin{array}{|c|c|c|}\hline 1&2&3\\\hline 4\\\cline{1-1}\end{array}\;\middle|\;\begin{array}{|c|c|c|}\hline 1&1&2\\\hline 3\\\cline{1-1}\end{array}\right)-\left(\begin{array}{|c|c|c|}\hline 1&2&3\\\hline 4\\\cline{1-1}\end{array}\;\middle|\;\begin{array}{|c|c|c|}\hline 3&1&2\\\hline 1\\\cline{1-1}\end{array}\right)\\&=-\left(\begin{array}{|c|c|c|}\hline 1&2&3\\\hline 4\\\cline{1-1}\end{array}\;\middle|\;\begin{array}{|c|c|c|}\hline 1&2&3\\\hline 1\\\cline{1-1}\end{array}\right)\end{aligned}$$

Example Straightening

Shuffle:

$$\left(\begin{array}{|c|c|}\hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \;\middle|\; \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) = \left(\begin{array}{|c|c|}\hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \;\middle|\; \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) - \left(\begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 & 4 \\ \hline \end{array} \;\middle|\; \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) + \left(\begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} \;\middle|\; \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right)$$

Laplace:

$$\begin{aligned} \left(\begin{array}{|c|c|}\hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \;\middle|\; \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array} \right) &= \left(\begin{array}{|c|c|c|}\hline 1 & 2 & 3 \\ \hline \multicolumn{1}{|c|}{4} \\ \cline{1-1} \end{array} \;\middle|\; \begin{array}{|c|c|c|}\hline 1 & 1 & 2 \\ \hline \multicolumn{1}{|c|}{3} \\ \cline{1-1} \end{array} \right) - \left(\begin{array}{|c|c|c|}\hline 1 & 2 & 3 \\ \hline \multicolumn{1}{|c|}{4} \\ \cline{1-1} \end{array} \;\middle|\; \begin{array}{|c|c|c|}\hline 3 & 1 & 2 \\ \hline \multicolumn{1}{|c|}{1} \\ \cline{1-1} \end{array} \right) \\ &= - \left(\begin{array}{|c|c|c|}\hline 1 & 2 & 3 \\ \hline \multicolumn{1}{|c|}{4} \\ \cline{1-1} \end{array} \;\middle|\; \begin{array}{|c|c|c|}\hline 1 & 2 & 3 \\ \hline \multicolumn{1}{|c|}{1} \\ \cline{1-1} \end{array} \right) \end{aligned}$$

We notice that our unstraightened rows appear in the first expression so that we can solve for it.

Example Straightening

$$\left(\begin{array}{|c|c|}\hline 2 & 3 \\ \hline 1 & 4 \\ \hline \end{array} \;\middle|\; \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array}\right) = \left(\begin{array}{|c|c|}\hline 1 & 3 \\ \hline 2 & 4 \\ \hline \end{array} \;\middle|\; \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array}\right) - \left(\begin{array}{|c|c|}\hline 1 & 2 \\ \hline 3 & 4 \\ \hline \end{array} \;\middle|\; \begin{array}{|c|c|}\hline 1 & 2 \\ \hline 1 & 3 \\ \hline \end{array}\right) - \left(\begin{array}{|c|c|c|}\hline 1 & 2 & 3 \\ \hline \multicolumn{1}{|c|}{4} \\ \cline{1-1} \end{array} \;\middle|\; \begin{array}{|c|c|c|}\hline 1 & 2 & 3 \\ \hline \multicolumn{1}{|c|}{1} \\ \cline{1-1} \end{array}\right)$$

Example Straightening

$$\left(\begin{ytableau} 2 & 3 \\ 1 & 4 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \end{ytableau}\right) = \left(\begin{ytableau} 1 & 3 \\ 2 & 4 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \end{ytableau}\right) - \left(\begin{ytableau} 1 & 2 \\ 3 & 4 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \end{ytableau}\right) - \left(\begin{ytableau} 1 & 2 & 3 \\ 4 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 & 3 \\ 1 \end{ytableau}\right)$$

Thus,

$$\left(\begin{ytableau} 2 & 3 \\ 1 & 4 \\ 2 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \\ 1 \end{ytableau}\right) = \left(\begin{ytableau} 1 & 3 \\ 2 & 4 \\ 2 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \\ 1 \end{ytableau}\right) - \left(\begin{ytableau} 1 & 2 \\ 3 & 4 \\ 2 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 \\ 1 & 3 \\ 1 \end{ytableau}\right) - \left(\begin{ytableau} 1 & 2 & 3 \\ 4 \\ 2 \end{ytableau} \;\middle|\; \begin{ytableau} 1 & 2 & 3 \\ 1 \\ 1 \end{ytableau}\right)$$

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All terms that appear have the same content and they are either the same shape with a lexicographically smaller filling or they have a longer shape.

Main Theorem

Conjecture ([Gro20, Conjecture 6.3])

Let $f \in I^{\det}_{n,n,n/2}$ be a nonzero polynomial. Then there is a constant depth f-oracle circuit of size $\mathrm{poly}(n)$ that computes the $t \times t$ determinant for some $t = \mathrm{poly}(n)$.

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Theorem ([DG25, Theorem 1.5])

Let $f \in I^{\det}_{n,m,r}$ be a nonzero polynomial. Then there is a depth-three f-oracle circuit of size $\mathrm{poly}(n, m, \deg(f))$ that *exactly computes* the $t \times t$ determinant for some $t = O\left(r^{1/3}\right)$.

Main Theorem

Theorem

Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Then, there exists a depth-three f-oracle circuit of size $O\left(n^2m^2d^3(n^3+m^3)\right)$ computing $(K_\sigma \mid K_\sigma)(X)$, where $\sigma_1 \geq r$ and $|\sigma| \leq d$.

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Definition (Canonical Bideterminant [dCEP80])

Let $\sigma = (4,2,1)$:

$$(K_\sigma \mid K_\sigma)(X) = \left(\begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & \multicolumn{2}{c}{} \\ \cline{1-2} 1 & \multicolumn{3}{c}{} \\ \cline{1-1} \end{array} \;\middle|\; \begin{array}{|c|c|c|c|} \hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & \multicolumn{2}{c}{} \\ \cline{1-2} 1 & \multicolumn{3}{c}{} \\ \cline{1-1} \end{array} \right)(X)$$

$$= \text{product of upper-left minors.}$$

Proof Idea

Since we are starting with $f(X) \in I^{\det}_{n,m,r} \subseteq \mathbb{F}[X]$

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Since we are starting with $f(X) \in I^{\det}_{n,m,r} \subseteq \mathbb{F}[X]$, we can express in terms of our standard basis:

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Proof Idea

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$$f(X) = \sum_{k \in I} c_k (S_k \mid T_k)(X).$$

Goal: Use $f(X)$ as an oracle to compute a canonical bideterminant.

Substitution

Definition (Substitution Sperator [dCEP80])

For $i < j$, the operator $\mathrm{Sub}_{i \to j}$ takes a tableau S and returns the tableau S' formed by taking each row of S which has an i but not a j, replacing that i with j, and sorting the row in increasing order.

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$$\underset{2\to 4}{\mathrm{Sub}} \begin{array}{|c|c|c|c|}\hline 1 & 2 & 3 & 4 \\ \hline 1 & 2 & \multicolumn{2}{c}{} \\ \cline{1-2} 1 & \multicolumn{3}{c}{} \\ \cline{1-1} \end{array} = \begin{array}{|c|c|c|c|}\hline 1 & 2 & 3 & 4 \\ \hline 1 & 4 & \multicolumn{2}{c}{} \\ \cline{1-2} 1 & \multicolumn{3}{c}{} \\ \cline{1-1} \end{array}$$

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Let $h_i^j(S)$ be the number of times i is replaced by j in S.

Substitution

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In fact, after enough applications we arrive at *anticanonical* tableau:

Let $\sigma = (4, 2, 1)$ and $n = m = 6$:

$$(\overline{K}_\sigma \mid \overline{K}_\sigma)(X) = \left(\begin{array}{|c|c|c|c|} \hline 3 & 4 & 5 & 6 \\ \hline 5 & 6 & \multicolumn{2}{c}{} \\ \cline{1-2} 6 & \multicolumn{3}{c}{} \\ \cline{1-1} \end{array} \;\middle|\; \begin{array}{|c|c|c|c|} \hline 3 & 4 & 5 & 6 \\ \hline 5 & 6 & \multicolumn{2}{c}{} \\ \cline{1-2} 6 & \multicolumn{3}{c}{} \\ \cline{1-1} \end{array} \right)(X)$$

$$= \text{product of lower-right minors.}$$

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In fact, after enough applications we arrive at *anticanonical* tableau:

Lemma ([dCEP80, stated before Corollary 1.7])

Let S be a conjugate semistandard tableau of shape σ such that S has entries of value at most n. Then

$$\Big(\underset{n-1\to n}{\mathrm{Sub}} \circ \underset{n-2\to n}{\mathrm{Sub}} \circ \underset{n-2\to n-1}{\mathrm{Sub}} \circ \cdots \circ \underset{2\to n}{\mathrm{Sub}} \circ \cdots \circ \underset{2\to 3}{\mathrm{Sub}} \circ \underset{1\to n}{\mathrm{Sub}} \circ \cdots \circ \underset{1\to 2}{\mathrm{Sub}}\Big)(S) = \overline{K}_\sigma.$$

Linear Operators

Substituting i for j corresponds to multiplication of X by an *elementary matrix* $E_{i,j}(\lambda)$, adding λ times row j to row i.

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Let λ be a new indeterminate and let $(S \mid T)$ be a bitableau.

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$$(S \mid T)(E_{i,j}(\lambda)X)$$

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Let λ be a new indeterminate and let $(S \mid T)$ be a bitableau.

$$(S \mid T)(E_{i,j}(\lambda)X) = \pm\lambda^{h_i^j(S)}(\operatorname*{Sub}_{i\to j}(S) \mid T)(X)$$

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Substituting i for j corresponds to multiplication of X by an *elementary matrix* $E_{i,j}(\lambda)$, adding λ times row j to row i.

Lemma

Let λ be a new indeterminate and let $(S \mid T)$ be a bitableau.

$$(S \mid T)(E_{i,j}(\lambda)X) = \pm\lambda^{h_i^j(S)}(\underset{i\to j}{\mathrm{Sub}}(S) \mid T)(X) + \sum_{h=0}^{h_i^j(S)-1} \lambda^h \sum_{S' \in \mathcal{C}_{i\to j}^h(S)} \pm(S' \mid T)(X).$$

For $0 \leq h \leq h_i^j(S) - 1$, let $\mathcal{C}_{i\to j}^h(S)$ be the set of tableaux of shape σ obtained by changing i to j at exactly h rows of S which contain i but not j and reordering those rows to be increasing.

Applying all Substitution Operators

Lemma ([dCEP80, stated before Corollary 1.7])

Let S be a conjugate semistandard tableau of shape σ such that S has entries of value at most n. Then

$$\Big(\underset{n-1\to n}{\mathrm{Sub}} \circ \underset{n-2\to n}{\mathrm{Sub}} \circ \underset{n-2\to n-1}{\mathrm{Sub}} \circ \cdots \circ \underset{2\to n}{\mathrm{Sub}} \circ \cdots \circ \underset{2\to 3}{\mathrm{Sub}} \circ \underset{1\to n}{\mathrm{Sub}} \circ \cdots \circ \underset{1\to 2}{\mathrm{Sub}}\Big)(S) = \overline{K}_{\sigma}.$$

Applying all Substitution Operators

Lemma ([dCEP80, stated before Corollary 1.7])

Let S be a conjugate semistandard tableau of shape σ such that S has entries of value at most n. Then

$$\Big(\underset{n-1\to n}{\mathrm{Sub}} \circ \underset{n-2\to n}{\mathrm{Sub}} \circ \underset{n-2\to n-1}{\mathrm{Sub}} \circ \cdots \\ \circ \underset{2\to n}{\mathrm{Sub}} \circ \cdots \circ \underset{2\to 3}{\mathrm{Sub}} \circ \underset{1\to n}{\mathrm{Sub}} \circ \cdots \circ \underset{1\to 2}{\mathrm{Sub}}\Big)(S) = \overline{K}_\sigma.$$

$$M = E_{1,2}(\lambda_{1,2}) \cdots E_{1,n}(\lambda_{1,n}) E_{2,3}(\lambda_{2,3}) \cdots E_{2,n}(\lambda_{2,n}) \\ \cdots E_{n-2,n-1}(\lambda_{n-2,n-1}) E_{n-2,n}(\lambda_{n-2,n}) E_{n-1,n}(\lambda_{n-1,n})$$

$$N = E_{m-1,m}(\xi_{m-1,m})^\top E_{m-2,m}(\xi_{m-2,m})^\top E_{m-2,m-1}(\xi_{m-2,m-1})^\top \\ \cdots E_{2,m}(\xi_{2,m})^\top \cdots E_{2,3}(\xi_{2,3})^\top E_{1,m}(\xi_{1,m})^\top \cdots E_{1,2}(\xi_{1,2})^\top.$$

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Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Then $f(MXN)$ can be expressed as a sum

$$f(MXN) = \sum_{k \in A} \widehat{c}_k \Lambda^{\overline{e}_k} \Xi^{\overline{f}_k} \cdot (\overline{K}_{\sigma_k} \mid \overline{K}_{\sigma_k})(X) + \sum_{\ell \in B} \widehat{c}_\ell \Lambda^{\overline{e}_\ell} \Xi^{\overline{f}_\ell} \cdot (S_\ell \mid T_\ell)(X)$$

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Lemma

$$(\overline{K}_\sigma \mid \overline{K}_\sigma)(J_n X J_m) = \pm (K_\sigma \mid K_\sigma)(X).$$

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Lemma ([DG25, Corollary 3.10])

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$$D = \mathrm{diag}(y_1 v, y_2 v^2 \ldots, y_n v^n) \in \mathbb{F}[Y, v]^{n \times n}$$
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Lemma

Let $(S \mid T)(X)$ be a bideterminant. Then

$$(S \mid T)(DXD') = Y^{\bar{s}} Z^{\bar{t}} v^{|S|+|T|} \cdot (S \mid T)(X).$$

- $(\bar{s}, \bar{t})$ is the multidegree of $(S \mid T)(X)$: $\mathrm{mdeg}(x_{i,j}) = e_i \oplus e_j$
- $|S|$ is the sum of the entries in S.

Isolating Terms

Lemma

Let $f(X) \in I^{\det}_{n,m,r}$ be a nonzero polynomial of degree d. Let $g = f(MJ_nDXD'J_mN) \in \mathbb{F}[X, \Lambda, \Xi, Y, Z, v]$.

$$\mathrm{coeff}_{v^{d_{\min}}}(g) = \sum_{k \in I} c_k \Lambda^{\overline{e}_k} \Xi^{\overline{f}_k} Y^{\overline{s}_k} Z^{\overline{t}_k} \cdot (K_{\sigma_k} \mid K_{\sigma_k})(X).$$

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We can compute g with a layer of Σ-gates followed by a single f-gate.

We now *interpolate* to get an f-oracle circuit computing $\mathrm{coeff}_{v^{d_{min}}}(g)$.

Interpolation

Definition

Consider a degree d polynomial $g(\overline{x}, t) = \sum_{i=0}^{d} \text{coeff}_{t^i}(g) t^i$.

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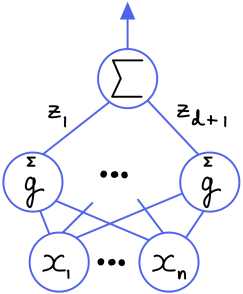

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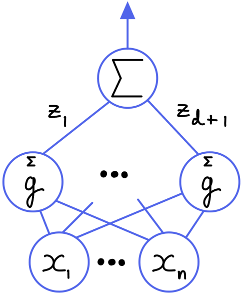

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Proof

Express *Lagrange interpolation* as a circuit.

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Problem: Interpolating with respect to multiple variables is *expensive*.

Issues with Interpolation

Setup: We want to *isolate* some coefficient $c_{\overline{e}}$ in $g(\overline{x}) = \sum_{\overline{e} \in \mathcal{L}} c_{\overline{e}} \overline{x}^{\overline{e}}$.

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Recall: Prior work [AF22] gave ideal closure results in the border setting.

If a circuit border computes $g(\overline{x})$, then it computes

$$g(\overline{x}) + \varepsilon^q \tilde{g}(\overline{x}, \varepsilon) \in \mathbb{F}[[\varepsilon]][\overline{x}], \qquad q \geq 1.$$

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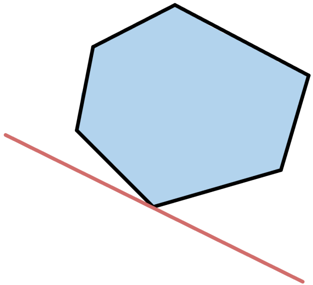
Linear programs with *random* cost functions will have a unique minimum.



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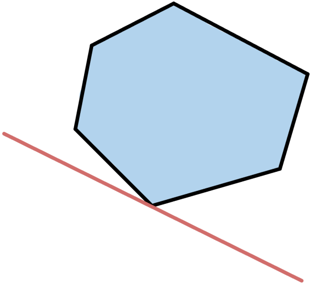

Moreover, if the linear equations have bounded integer coefficients, then evaluation at *small, random* values has a unique minimum.

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Let $\mathcal{L}$ be any collection of distinct linear forms in variables $z_1, \ldots, z_\ell$ with coefficients in the range $\{ 0, \ldots, K \}$ for some integer $K \in \mathbb{Z}_{\geq 0}$. Let $\varepsilon > 0$.

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Then, with probability at least $1 - \varepsilon$, there is a unique form in $\mathcal{L}$ of minimum value at $z_1, \ldots, z_\ell$.

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Lemma ([DG25, Corollary 2.27])

Consider a polynomial $g(x_1, \ldots, x_\ell)$ such that the individual degree of each x_i in g is at most K:

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Apply the *Isolation Lemma* to get an f-oracle circuit computing some

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Theorem ([DG25, Theorem 1.6])

Let $f \in I_{2n,2r}^{\text{pfaff}}$ be a nonzero polynomial. Then there is a depth-three f-oracle circuit of size $\text{poly}(n, \deg(f))$ that *exactly computes* the $t \times t$ Pfaffian for some $t = O\left(r^{1/3}\right)$.

Symmetric Analogue?

Let $x_{i,j}$ be indeterminates for $1 \leq i \leq j \leq n$ and let $x_{j,i} \coloneqq x_{i,j}$ for $1 \leq j < i \leq n$. Thus, $X = (x_{i,j})_{1 \leq i,j \leq n}$ is a symmetric $n \times n$ matrix.

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Conjecture

Let $I_{n,r}^{\mathrm{sym}} \subseteq \mathbb{F}[X]$ be the ideal generated by the $r \times r$ minors of X. Let $f \in I_{n,r}^{\mathrm{sym}}$ be a nonzero polynomial. Then there exists a constant-depth f-oracle circuit of size $\mathrm{poly}(n, \deg(f))$ that computes the $t \times t$ determinant of a symmetric matrix for $t = \mathrm{poly}(n)$.

Partial Progress

Theorem

Let $f(X) \in I_{n,r}^{\text{sym}}$ be a nonzero polynomial of degree d. Then, there exists a depth-three f-oracle circuit of size $O\left(d^3 n^7\right)$ computing $(K_\sigma \mid K_\sigma)(X)$, where $\sigma_1 \geq r$ and $|\sigma| \leq d$.

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Our main roadblock for fully resolving the symmetric case is the reduction from a product of determinants to a single determinant.

Roadblocks for the Permanent

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The main roadblock is that we have no analogue of the following result:

$$(S \mid T)(E_{i,j}(\lambda)X) = \pm\lambda^{h_i^j(S)}(\underset{i\to j}{\mathrm{Sub}}(S) \mid T)(X) + \sum_{h=0}^{h_i^j(S)-1} \lambda^h \sum_{S' \in \mathcal{C}_{i\to j}^h(S)} \pm(S' \mid T)(X).$$

Algebras with Straightening Laws / Hodge Algebras

Definition ([Eis80, dCEP82])

Let A be a ring, $H \subseteq A$ is a poset.

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Let R be a ring, A an R-algebra, and $H \subseteq A$ a finite poset which generates A as an R-algebra. A is an *Algebra with a Straightening Law* if

- A is a free R-module with a basis given by the standard monomials
- if $\alpha, \beta \in H$ are incomparable, and if

$$\alpha\beta = \sum_{i=1}^{m} r_i \gamma_{i,1} \cdots \gamma_{i,k_i}$$

  where $0 \neq r_i \in R$ and each $\gamma_{i,1} \cdots \gamma_{i,k_i}$ is a standard monomial, we must have that for all $1 \leq i \leq m$ that $\gamma_{i,1} \leq \alpha, \beta$.

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These relations in the second requirement are known as the *straightening relations* or *straightening laws*.

Hankel Ideals

Let $\mathbb{F}$ be a field. Let x_i be indeterminates for $1 \leq i \leq n$. For $1 \leq j \leq n$, let X_j be the following Hankel matrix:

$$\begin{pmatrix} x_1 & x_2 & x_3 & \cdots & x_{n-j+1} \\ x_2 & x_3 & x_4 & \cdots & x_{n-j+2} \\ x_3 & x_4 & x_5 & \cdots & x_{n-j+3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_j & x_{j+1} & x_{j+2} & \cdots & x_n \end{pmatrix}.$$

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For $1 \leq r \leq \min\{\, j, n-j+1 \,\}$ let $I_r(X_j)$ be the ideal generated by $r \times r$ minors of X_j.

Hankel Ideals

The ideals $I_r(X_j)$ do not have the structure of an algebra with a straightening law [Con98].

In particular, the product of standard monomials need not be standard.

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In particular, the product of standard monomials need not be standard.

We still conjecture the following:

Conjecture

Let $I_r(X_j) \subseteq \mathbb{F}[x_1, \ldots, x_n]$ be the ideal generated by the $r \times r$ minors of X_j. Let $f \in I_r(X_j)$ be a nonzero polynomial. Then there exists a constant-depth f-oracle circuit of size $\mathrm{poly}(n, \deg(f))$ that computes the $t \times t$ determinant of a Hankel matrix for $t = \mathrm{poly}(n)$.

Failure of Prior Strategy

In the general determinantal, skew-symmetric, and symmetric cases we analyzed the behavior of $E_{i,j}(\lambda)$.

Failure of Prior Strategy

In the general determinantal, skew-symmetric, and symmetric cases we analyzed the behavior of $E_{i,j}(\lambda)$.

This no longer works:

$$\begin{pmatrix} 1 & \lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 & x_2 & x_3 \\ x_2 & x_3 & x_4 \\ x_3 & x_4 & x_5 \end{pmatrix} = \begin{pmatrix} x_1 + \lambda x_2 & x_2 + \lambda x_3 & x_3 + \lambda x_4 \\ x_2 & x_3 & x_4 \\ x_3 & x_4 & x_5 \end{pmatrix}$$

Thank You!

If you don't know how to answer someone's question, you should compliment them by saying it's a good question.

— Noga Alon

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